

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Concept Mapping

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	DHIVYADHARSAN M.S	Reg. No.:	192511465
Section / Batch:	2025	Date:	10/07/2026

Code of Conduct

I DHIVYADHARSAN M.S. _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: DHARSAN _____

Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	2	20
Search Data Structure	Linear Search and Binary Search	3	20
Recursion, Performance analysis	Recursion	4	20
Tree Data Structures	Classifications of Tree data structure	5	20
GRAND TOTAL:		100 Marks	

Annexure A – Concept Mapping

1. A student attendance system records daily presence data. Construct a concept map linking Array → Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each structure works and justify which is more suitable for dynamic data updates. (20 Marks)
2. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing. (20 Marks)
3. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)
4. A program processes nested expressions using recursion and may encounter deep nesting. Construct a concept map linking Recursion → Call Stack → Stack Overflow → Error Handling → Efficiency. Analyze potential issues and justify strategies to handle them. (20 Marks)
5. A company maintains hierarchical employee records with multiple levels of management. Construct a concept map connecting Tree → Nodes → Levels → Traversal (DFS/BFS) → Searching → Efficiency. Analyze how traversal techniques affect performance and justify the most suitable approach for retrieving employee data. (20 Marks)

Rubrics for Calculation

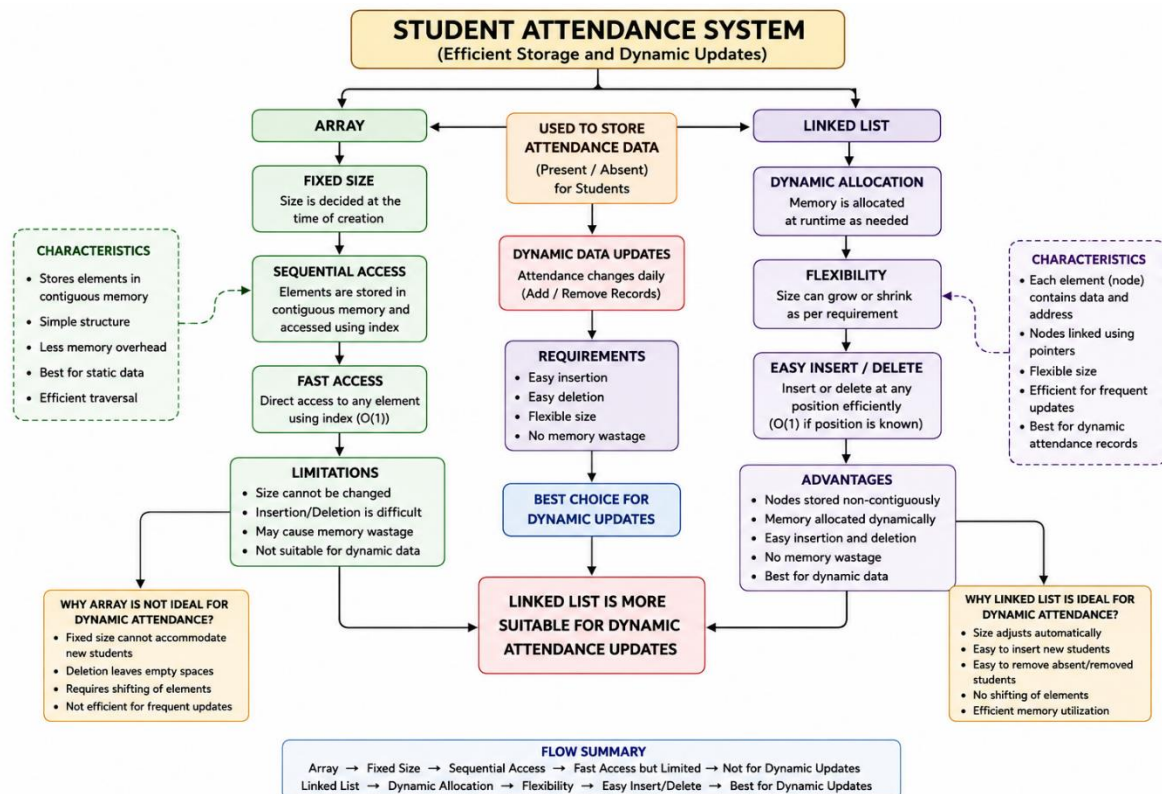
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CSA0305- DATA STRUCTURE CONCEPT MAPPING

By
M.S.DHIVYADHARSAN
(192511465)

1. A student attendance system records daily presence data. Construct a concept map linking Array → Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each structure works and justify which is more suitable for dynamic data updates.



```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct Node
```

```
{
```

```
    int rollNo;
```

```
    char status;
```

```
    struct Node *next;
```

```
};
```

```
int main()
{
    struct Node *head = NULL, *temp, *newNode;
    int n, i;

    printf("Enter number of students: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++)
    {
        newNode = (struct Node*)malloc(sizeof(struct Node));

        printf("Enter Roll No: ");
        scanf("%d", &newNode->rollNo);

        printf("Enter Status (P/A): ");
        scanf(" %c", &newNode->status);

        newNode->next = NULL;

        if(head == NULL)
            head = temp = newNode;
        else
        {
            temp->next = newNode;
        }
    }
}
```

```

        temp = newNode;
    }
}

printf("\nAttendance List\n");

temp = head;

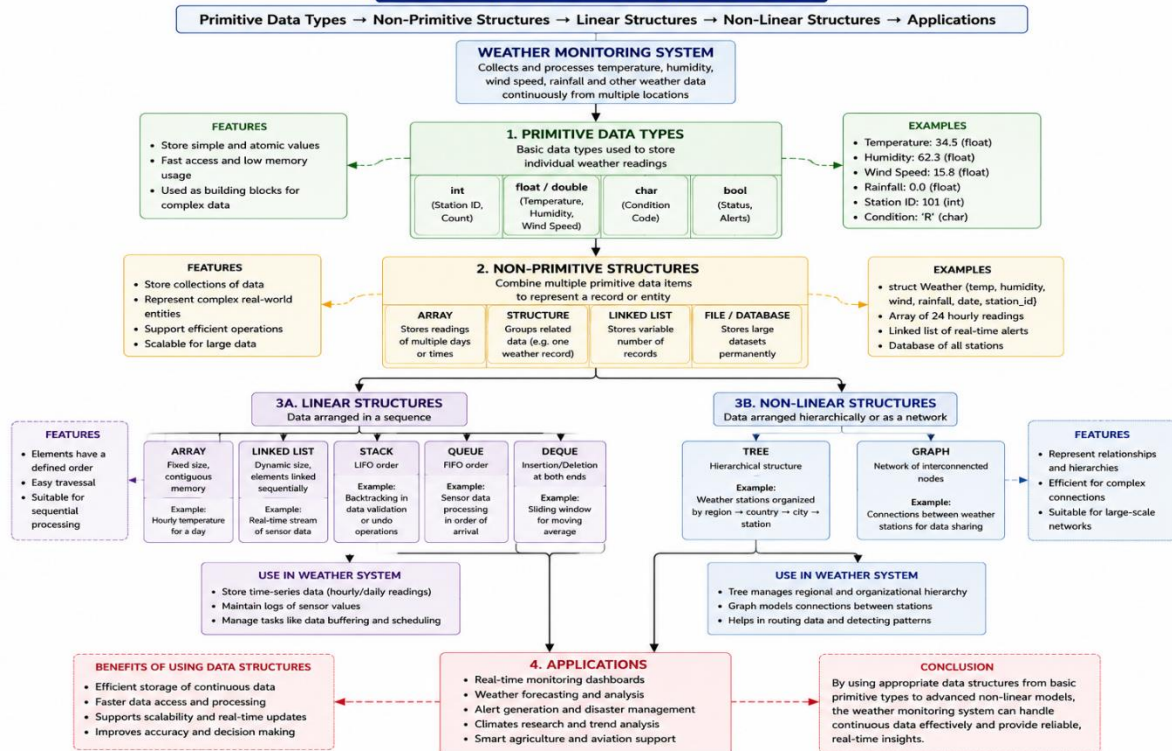
while(temp != NULL)
{
    printf("Roll No: %d  Status: %c\n", temp->rollNo, temp-
>status);
    temp = temp->next;
}

return 0;
}

```

2. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing.

Q2. WEATHER MONITORING SYSTEM



```
#include <stdio.h>
```

```
struct Weather
```

```
{
```

```
    float temperature;
```

```
    float humidity;
```

```
    float windSpeed;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Weather data[5];
```

```
    int i;
```

```
for(i=0;i<5;i++)
{
    printf("Station %d\n",i+1);

    printf("Temperature: ");
    scanf("%f",&data[i].temperature);

    printf("Humidity: ");
    scanf("%f",&data[i].humidity);

    printf("Wind Speed: ");
    scanf("%f",&data[i].windSpeed);
}
```

```
printf("\nWeather Report\n");
```

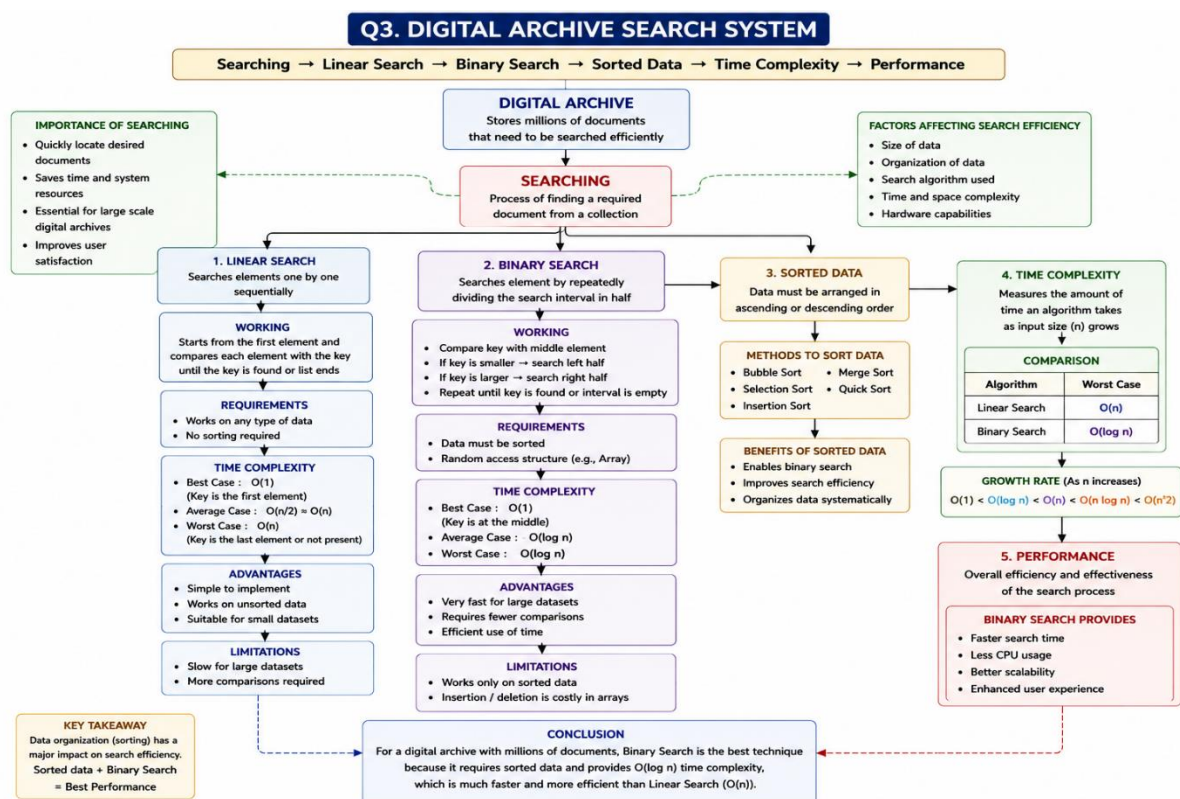
```
for(i=0;i<5;i++)
{
    printf("Station %d : %.2f %.2f %.2f\n",
    i+1,
    data[i].temperature,
    data[i].humidity,
    data[i].windSpeed);
}
```



```
return 0;
```

```
}
```

3. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique.



```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int a[20], n, key;
```

```
int low = 0, high, mid, i;
```

```
printf("Enter number of documents: ");  
scanf("%d",&n);
```

```
printf("Enter sorted document IDs:\n");
```

```
for(i=0;i<n;i++)  
    scanf("%d",&a[i]);
```

```
printf("Enter document ID to search: ");  
scanf("%d",&key);
```

```
high = n-1;
```

```
while(low<=high)  
{  
    mid=(low+high)/2;
```

```
    if(a[mid]==key)  
    {  
        printf("Document Found");  
        return 0;  
    }
```

```
    else if(a[mid]<key)  
        low=mid+1;
```

```

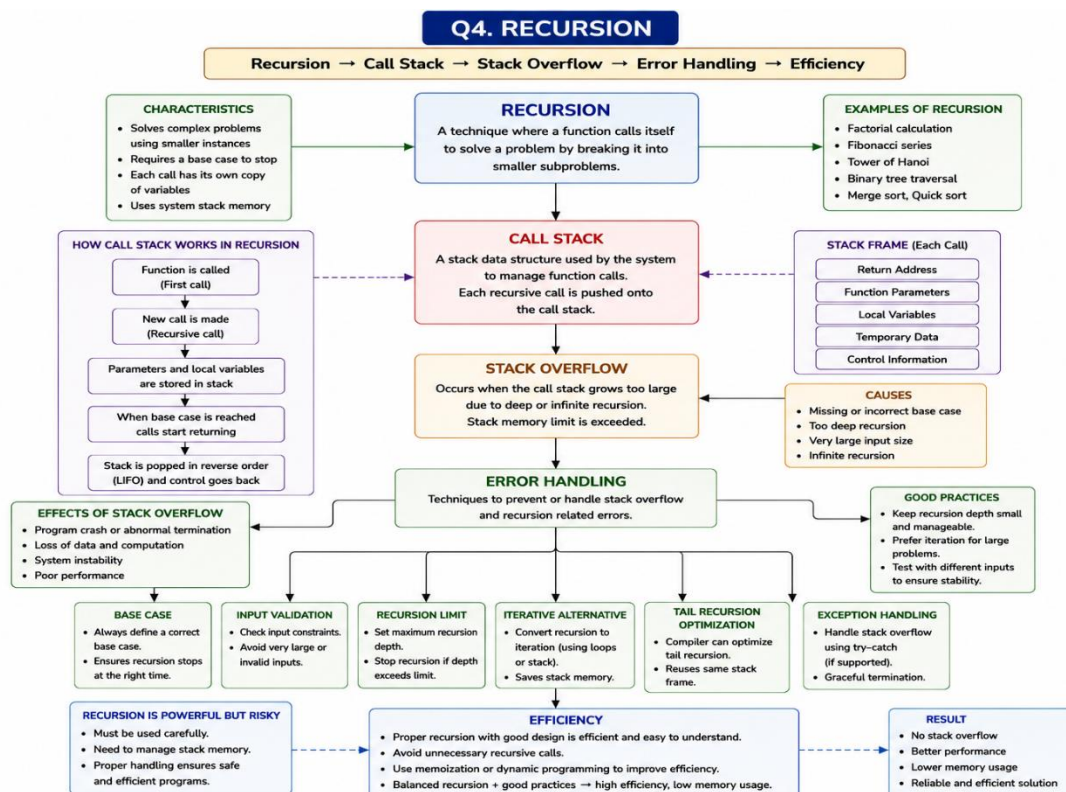
else
    high=mid-1;
}

printf("Document Not Found");

return 0;
}

```

4. A program processes nested expressions using recursion and may encounter deep nesting. Construct a concept map linking Recursion → Call Stack → Stack Overflow → Error Handling → Efficiency. Analyze potential issues and justify strategies to handle them.



```
#include <stdio.h>
```

```
long long factorial(int n)
{
    if(n==0 || n==1)
        return 1;

    return n * factorial(n-1);
}
```

```
int main()
{
    int n;

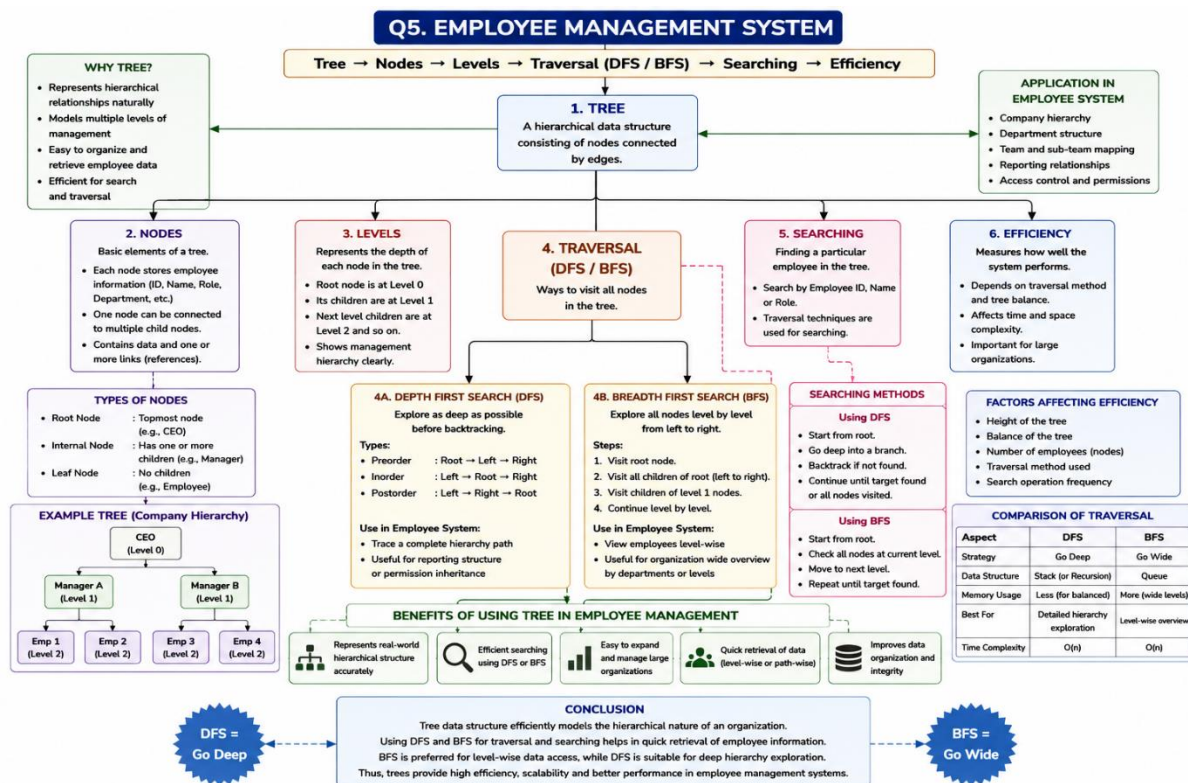
    printf("Enter a number: ");
    scanf("%d",&n);

    if(n<0)
        printf("Invalid Input");
    else
        printf("Factorial = %lld",factorial(n));

    return 0;
}
```

5. A company maintains hierarchical employee records with multiple levels of management. Construct a concept map

connecting Tree → Nodes → Levels → Traversal (DFS/BFS) → Searching → Efficiency. Analyze how traversal techniques affect performance and justify the most suitable approach for retrieving employee data.



```
#include <stdio.h>
```

```
struct Employee
{
    int id;
    char name[30];
    int parent;
};
```

```
int main()
{
```

```
struct Employee emp[10];
int n,i;

printf("Enter number of employees: ");
scanf("%d",&n);

for(i=0;i<n;i++)
{
    printf("\nEmployee %d\n",i+1);

    printf("ID: ");
    scanf("%d",&emp[i].id);

    printf("Name: ");
    scanf("%s",emp[i].name);

    printf("Manager ID (-1 for CEO): ");
    scanf("%d",&emp[i].parent);
}

printf("\nEmployee Hierarchy\n");

for(i=0;i<n;i++)
{
    printf("Employee ID: %d\tName: %s\tManager ID: %d\n",
```

```
    emp[i].id,  
    emp[i].name,  
    emp[i].parent);  
}  
  
return 0;  
}
```